

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 12B

LATTICE ENERGY, BORN-HABER CYCLES & SOLUTION ENERGETICS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 12: ENTROPY AND ENERGETICS
SUB-TOPIC:	12B: Lattice Energy, Born-Haber Cycles & Solution Energetics
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	233 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25

Q1. 1. Lattice Energy Definition

[3 Marks]
WCH14/01/Jan23/Q7

Lattice energy is a measure of the strength of ionic bonding in a crystal lattice.

(a) Define standard lattice energy ($\Delta_{LE} H^\circ$). [2]

(b) Explain why lattice energy values are always negative. [1]

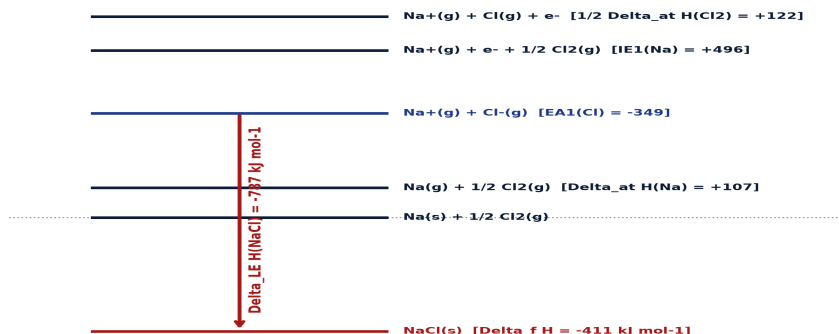
Q2. 2. Born-Haber Cycle Calculation for NaCl

[5 Marks]
WCH14/01/Oct22/Q8

Consider the following standard enthalpy changes (kJ mol^{-1}):

$\Delta_f H(\text{NaCl}) = -411$, $\Delta_{at} H(\text{Na}) = +107$, $IE_1(\text{Na}) = +496$, $\frac{1}{2} \Delta_{at} H(\text{Cl}_2) = +122$, $EA_1(\text{Cl}) = -349$.

Born-Haber Cycle for Sodium Chloride, NaCl(s)



(a) Construct the Born-Haber cycle for NaCl(s) and calculate the lattice energy $\Delta_{LE} H(\text{NaCl})$. [4]

(b) State how lattice energy relates to the melting point of NaCl. [1]

Q3. 3. First Electron Affinity Definition & Sign**[3 Marks]**
WCH14/01/Jun22/Q9

The first electron affinity of chlorine is $EA1 = -349 \text{ kJ mol}^{-1}$.

- (a) Define first electron affinity. [2]

- (b) Explain why the first electron affinity of chlorine is exothermic. [1]

Q4. 4. Second Electron Affinity Endothermic Nature**[3 Marks]**
WCH14/01/Jan22/Q10

The second electron affinity of oxygen is $EA2(O) = +798 \text{ kJ mol}^{-1}$.

- (a) Write an equation representing the second electron affinity of oxygen. [1]

- (b) Explain why the second electron affinity is endothermic while the first is exothermic. [2]

Q5. 5. Lattice Energy Comparison: NaCl vs MgCl₂**[4 Marks]**
WCH14/01/Oct21/Q8

$\Delta_{LE} H(\text{NaCl}) = -787 \text{ kJ mol}^{-1}$, whereas $\Delta_{LE} H(\text{MgCl}_2) = -2525 \text{ kJ mol}^{-1}$.

- (a) Explain in terms of ionic charge and ionic radius why MgCl_2 has a significantly more exothermic lattice energy than NaCl . [4]

**Q6. 6. Experimental vs Theoretical Lattice Energies:
Polarization & Covalency****[5 Marks]**
WCH14/01/Jun21/Q9

For AgI: Experimental Born-Haber lattice energy = -889 kJ mol^{-1} ; Theoretical ionic model lattice energy = -758 kJ mol^{-1} .

- (a) Explain why the experimental lattice energy is significantly more exothermic than the theoretical value. [4]

- (b) State which ion acts as the polarizing cation and which as the polarizable anion. [1]

Q7. 7. Fajans' Rules for Ionic Polarization**[4 Marks]**
WCH14/01/Jan21/Q10

Fajans' rules summarize factors that favour covalent character in ionic compounds.

- (a) State the two properties of a cation and two properties of an anion that maximize polarization. [4]

Q8. 8. Enthalpy Change of Solution & Hydration Cycle**[5 Marks]**
WCH14/01/Oct20/Q8

For MgSO_4 : $\Delta_{\text{LE}} H = -2870 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H(\text{Mg}^{2+}) = -1920 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H(\text{SO}_4^{2-}) = -1041 \text{ kJ mol}^{-1}$.

- (a) Construct an enthalpy cycle relating $\Delta_{\text{sol}} H$, $\Delta_{\text{LE}} H$, and $\Delta_{\text{hyd}} H$. [2]

(b) Calculate $\Delta_{\text{sol}} H$ for MgSO_4 . [3]

Q9. 9. Solubility Trend of Group 2 Sulfates

[5 Marks]

WCH14/01/Jun20/Q10

Solubility of Group 2 sulfates decreases down the group (MgSO_4 soluble $\rightarrow$ BaSO_4 insoluble).

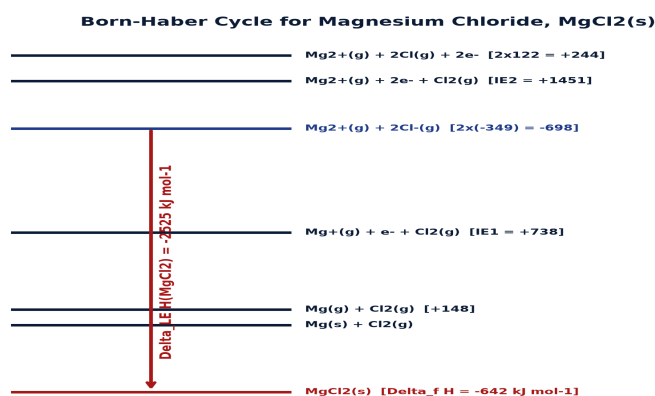
(a) Explain this trend in terms of changes in lattice energy and hydration enthalpy down the group. [5]

Q10. 10. Born-Haber Cycle Calculation for MgCl_2

[5 Marks]

WCH14/01/Jan20/Q10

Data (kJ mol^{-1}): $\Delta_{\text{f}} H(\text{MgCl}_2) = -642$, $\Delta_{\text{at}} H(\text{Mg}) = +148$, $\text{IE}_1(\text{Mg}) = +738$, $\text{IE}_2(\text{Mg}) = +1451$, $2 \times \frac{1}{2} \Delta_{\text{at}} H(\text{Cl}_2) = +244$, $2 \times \text{EA}_1(\text{Cl}) = -698$.



(a) Calculate the lattice energy $\Delta_{\text{LE}} H$ for $\text{MgCl}_2(\text{s})$. [4]

(b) Compare $\Delta_{\text{LE}} H(\text{MgCl}_2)$ with $\Delta_{\text{LE}} H(\text{NaCl})$ (-787 kJ mol^{-1}). [1]

Q11. 11. Solubility Trend of Group 2 Hydroxides

[5 Marks]
WCH14/01/Oct19/Q11

Solubility of Group 2 hydroxides increases down the group ($\text{Mg}(\text{OH})_2$ insoluble $\rightarrow$ $\text{Ba}(\text{OH})_2$ soluble).

(a) Explain why the solubility trend of hydroxides is opposite to that of sulfates. [5]

Q12. 12. Enthalpy of Hydration Definition & Trend

[4 Marks]
WCH14/01/Jun19/Q12

$\Delta_{\text{hyd}} H^\circ$ values (kJ mol^{-1}): $\text{Na}^+ = -406$, $\text{Mg}^{2+} = -1920$, $\text{Al}^{3+} = -4665$.

(a) Define standard enthalpy change of hydration. [2]

(b) Explain the trend in $\Delta_{\text{hyd}} H^\circ$ across Na^+ , Mg^{2+} , and Al^{3+} . [2]

[3 Marks]

WCH14/01/Jan19/Q13

Q13. 13. Enthalpy of Atomisation Definition

For chlorine: $\frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$ ($\Delta_{\text{at}} H = +122 \text{ kJ mol}^{-1}$).

(a) Define standard enthalpy change of atomisation. [2]

(b) Relate $\Delta_{\text{at}} H$ of chlorine to its bond dissociation enthalpy. [1]

[4 Marks]

WCH14/01/Sample/Q14

Q14. 14. Theoretical Lattice Energy Calculation via Born-Landé Equation

The ionic model assumes ions are perfectly spherical with purely electrostatic attraction.

(a) State the two assumptions of the purely ionic model. [2]

(b) Explain why experimental lattice energy matches theoretical value for NaCl but not for AgCl. [2]

[5 Marks]

WCH14/01/sample/Q15

Q15. 15. Born-Haber Cycle for Calcium Fluoride CaF_2

Data (kJ mol^{-1}): $\Delta_{\text{f}} H(\text{CaF}_2) = -1220$, $\Delta_{\text{at}} H(\text{Ca}) = +178$, $\text{IE}_1(\text{Ca}) = +590$, $\text{IE}_2(\text{Ca}) = +1145$, $2 \times \frac{1}{2} \Delta_{\text{at}} H(\text{F}_2) = +158$, $2 \times \text{EA}_1(\text{F}) = -656$.

(a) Calculate $\Delta_{\text{LE}} H$ for CaF_2 . [4]

(b) State why 2 moles of F atoms undergo electron affinity. [1]

[4 Marks]

Q16. 16. Enthalpy of Solution of Sodium Hydroxide

WCH14/01/Sample/Q16

$\text{NaOH(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$: $\Delta_{\text{LEH}} = -900 \text{ kJ mol}^{-1}$, $\Delta_{\text{hydH}}(\text{Na}^+) = -406 \text{ kJ mol}^{-1}$, $\Delta_{\text{hydH}}(\text{OH}^-) = -539 \text{ kJ mol}^{-1}$.

(a) Calculate Δ_{solH} for NaOH. [3]

(b) Explain why dissolving NaOH in water causes a temperature rise. [1]

[5 Marks]

Q17. 17. Born-Haber Cycle for Potassium Oxide K₂O

WCH14/01/Sample/Q17

Data (kJ mol^{-1}): $\Delta_{\text{fH}}(\text{K}_2\text{O}) = -363$, $2 \times \Delta_{\text{atH}}(\text{K}) = +178$, $2 \times \text{IE1}(\text{K}) = +836$, $\frac{1}{2} \Delta_{\text{atH}}(\text{O}_2) = +249$, $\text{EA1}(\text{O}) = -141$, $\text{EA2}(\text{O}) = +798$.

(a) Calculate Δ_{LEH} for $\text{K}_2\text{O(s)}$. [5]

[3 Marks]

Q18. 18. Polarizing Power Comparison: Li⁺ vs K⁺

WCH14/01/sample/Q18

LiI has experimental lattice energy -849 kJ mol^{-1} (theoretical -738), whereas KI is -649 (theoretical -637).

(a) Explain why LiI has much greater covalent character than KI. [3]

[4 Marks]

Q19. 19. Enthalpy of Hydration of Halide Ions

WCH14/01/sample/Q19

Delta_{hyd} H° (kJ mol⁻¹): F⁻ = -506 , Cl⁻ = -364 , Br⁻ = -335 , I⁻ = -293 .

(a) Explain the trend in hydration enthalpy down Group 17. [4]

[5 Marks]

Q20. 20. Born-Haber Cycle for Aluminum Oxide Al₂O₃

WCH14/01/sample/Q20

Al₂O₃ has an exceptionally large exothermic lattice energy (Delta_{LE} H = $-15116 \text{ kJ mol}^{-1}$).

(a) Explain why Al₂O₃ has a vastly more exothermic lattice energy than NaCl (-787 kJ mol^{-1}). [5]

[4 Marks]

Q21. 21. Thermal Decomposition of Group 2 Nitrates

WCH14/01/sample/Q21

$2\text{Mg}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MgO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$. Decomposition temperature increases down Group 2.

(a) Explain why $\text{Mg}(\text{NO}_3)_2$ decomposes at a lower temperature than $\text{Ba}(\text{NO}_3)_2$. [4]

[4 Marks]

Q22. 22. Solubility of Calcium Hydroxide vs Calcium Sulfate

WCH14/01/sample/Q22

$\text{Ca}(\text{OH})_2$ is sparingly soluble (pH ~12), while CaSO_4 is poorly soluble.

(a) Relate their lattice energies and hydration enthalpies to explain why $\text{Ca}(\text{OH})_2$ is more soluble than CaSO_4 . [4]

[5 Marks]

Q23. 23. Born-Haber Cycle for Lithium Iodide LiI

WCH14/01/sample/Q23

Data (kJ mol^{-1}): $\Delta_f H(\text{LiI}) = -270$, $\Delta_{\text{at}} H(\text{Li}) = +159$, $\text{IE}(\text{Li}) = +520$, $\frac{1}{2} \Delta_{\text{at}} H(\text{I}_2) = +107$, $\text{EA}(\text{I}) = -295$.

(a) Calculate $\Delta_{\text{LE}} H$ for $\text{LiI}(\text{s})$. [4]

(b) Explain why LiI is soluble in organic solvents like ethanol. [1]

Q24. 24. Enthalpy of Hydration vs Lattice Energy for BaCl₂

[4 Marks]

WCH14/01/Sample/Q24

$\text{BaCl}_2(\text{s}) \rightarrow \text{Ba}^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$: $\Delta_{\text{LE}} H = -2056 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H(\text{Ba}^{2+}) = -1305 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H(\text{Cl}^{-}) = -364 \text{ kJ mol}^{-1}$.

(a) Calculate $\Delta_{\text{sol}} H$ for BaCl₂. [3]

(b) State whether dissolution is exothermic or endothermic. [1]

Q25. 25. Born-Haber Cycle for Copper(II) Oxide CuO

[5 Marks]

WCH14/01/Sample/Q25

Data (kJ mol^{-1}): $\Delta_{\text{f}} H(\text{CuO}) = -157$, $\Delta_{\text{at}} H(\text{Cu}) = +338$, $\text{IE}_1(\text{Cu}) = +746$, $\text{IE}_2(\text{Cu}) = +1958$, $\frac{1}{2} \Delta_{\text{at}} H(\text{O}_2) = +249$, $\text{EA}_1(\text{O}) + \text{EA}_2(\text{O}) = +657$.

(a) Calculate $\Delta_{\text{LE}} H$ for CuO(s). [5]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50

Q26. 26. A * Challenge: Born-Haber Cycle with Unknown Electron Affinity

[5 Marks]
WCH14/01/Hard/Q26

For metal bromide MBr_2 : $\Delta_f H = -524 \text{ kJ mol}^{-1}$, $\Delta_{\text{at}} H(\text{M}) = +178$, $\text{IE}_1(\text{M}) = +590$, $\text{IE}_2(\text{M}) = +1145$, $\Delta_{\text{at}} H(\text{Br}_2) = +112$, $\Delta_{\text{LE}} H = -2150$.

(a) Calculate the first electron affinity of bromine, $EA_1(\text{Br})$, in kJ mol^{-1} . [5]

Q27. 27. A* Challenge: Percent Covalent Character from Lattice Energy Comparison

[5 Marks]
WCH14/01/Hard/Q27

For AgBr: Experimental Born-Haber lattice energy = -905 kJ mol^{-1} ; Theoretical ionic model lattice energy = -758 kJ mol^{-1} .

(a) Calculate the percentage difference between experimental and theoretical lattice energy. [2]

(b) Explain the structural origin of this discrepancy in detail. [3]

Q28. 28. A* Challenge: Enthalpy of Solution Temperature Dependence & Entropy

[5 Marks]
WCH14/01/Hard/Q28

For KI: $\Delta_{LE} H = -649 \text{ kJ mol}^{-1}$, $\Delta_{hyd} H(K^+) = -322$, $\Delta_{hyd} H(I^-) = -293$.

(a) Calculate $\Delta_{\text{sol}} H$ for KI. [2]

(b) Predict how the solubility of KI changes with increasing temperature using Le Chatelier and entropy. [3]

Q29. 29. A* Challenge: Lattice Energy of Hypothetical NaCl₂

[6 Marks]
WCH14/01/Hard/Q29

Consider why NaCl₂(s) does not form despite its high theoretical lattice energy.

(a) Estimate $\Delta_{\text{LE}} H$ for NaCl₂(s) assuming it is similar to MgCl₂ (−2500 kJ mol^{−1}). [1]

(b) Explain why the second ionisation energy of sodium ($IE_2 = +4562 \text{ kJ mol}^{-1}$) makes $\Delta_f H(\text{NaCl}_2)$ highly endothermic. [3]

(c) Conclude why NaCl(s) forms instead of NaCl₂(s). [2]

Q30. 30. A* Challenge: Hydration Enthalpy vs Lattice Energy Competition in Halides

[5 Marks]
WCH14/01/Hard/Q30

Enthalpies of solution $\Delta_{\text{sol}} H$ (kJ mol^{−1}): LiF = +4.7, LiCl = −37.0, LiBr = −48.8, LiI = −63.3.

(a) Explain the trend in $\Delta_{\text{sol}} H$ down the lithium halides. [5]

**Q31. 31. A* Challenge: Born-Haber Cycle for Calcium Oxide
CaO****[5 Marks]**
WCH14/01/Hard/Q31

Data (kJ mol^{-1}): $\Delta_f H(\text{CaO}) = -635$, $\Delta_{\text{at}} H(\text{Ca}) = +178$, $\text{IE}_1(\text{Ca}) = +590$, $\text{IE}_2(\text{Ca}) = +1145$, $\frac{1}{2} \Delta_{\text{at}} H(\text{O}_2) = +249$, $\text{EA}_1(\text{O}) = -141$, $\text{EA}_2(\text{O}) = +798$.

(a) Calculate $\Delta_{\text{LE}} H$ for $\text{CaO}(\text{s})$. [4]

(b) Compare $\Delta_{\text{LE}} H(\text{CaO})$ with $\Delta_{\text{LE}} H(\text{NaCl})$ (-787 kJ mol^{-1}). [1]

**Q32. 32. A* Challenge: Polarization Effect on Thermal Stability
of Carbonates****[5 Marks]**
WCH14/01/Hard/Q32

Compare thermal decomposition of Na_2CO_3 (does not decompose at 800°C) vs CaCO_3 (decomposes at 840°C) vs MgCO_3 (decomposes at 300°C).

(a) Explain why Na^+ causes less polarization than Ca^{2+} and Mg^{2+} . [3]

(b) Explain why MgCO_3 decomposes at a much lower temperature than CaCO_3 . [2]

Q33. 33. A* Challenge: Born-Haber Cycle for Barium Chloride
BaCl₂

[5 Marks]
WCH14/01/Hard/Q33

Data (kJ mol^{-1}): $\Delta_f H(\text{BaCl}_2) = -858$, $\Delta_{\text{at}} H(\text{Ba}) = +180$, $\text{IE}_1(\text{Ba}) = +503$, $\text{IE}_2(\text{Ba}) = +965$, $2 \times \frac{1}{2} \Delta_{\text{at}} H(\text{Cl}_2) = +244$, $2 \times \text{EA}_1(\text{Cl}) = -698$.

(a) Calculate $\Delta_{\text{LE}} H$ for $\text{BaCl}_2(\text{s})$. [4]

(b) Compare $\Delta_{\text{LE}} H(\text{BaCl}_2)$ with $\Delta_{\text{LE}} H(\text{MgCl}_2)$ ($-2525 \text{ kJ mol}^{-1}$). [1]

Q34. 34. A* Challenge: Hydration Enthalpy vs Ion Size in
Transition Metal Ions

[5 Marks]
WCH14/01/Hard/Q34

$\Delta_{\text{hyd}} H^\circ$ (kJ mol^{-1}): $\text{Fe}^{2+} = -1946$, $\text{Fe}^{3+} = -4430$.

(a) Explain why Fe^{3+} has a vastly more exothermic hydration enthalpy than Fe^{2+} . [3]

(b) Predict how this affects the acidity of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ vs $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ solutions. [2]

Q35. 35. A* Challenge: Enthalpy Cycle for Anhydrous vs Hydrated Copper Sulfate**[5 Marks]**
WCH14/01/Hard/Q35

$\text{CuSO}_4(\text{s}) + 5\text{H}_2\text{O}(\text{l}) \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$. $\Delta_{\text{sol}} H(\text{CuSO}_4(\text{s})) = -66.5 \text{ kJ mol}^{-1}$; $\Delta_{\text{sol}} H(\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})) = +11.7 \text{ kJ mol}^{-1}$.

(a) Construct a Hess's law cycle to calculate $\Delta H_{\text{hydration}}$ for $\text{CuSO}_4(\text{s}) \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$. [4]

(b) Why cannot $\Delta H_{\text{hydration}}$ be measured directly by calorimetry? [1]

Q36. 36. A* Challenge: Enthalpy of Solution of Potassium Nitrate KNO₃**[5 Marks]**
WCH14/01/Hard/Q36

$\text{KNO}_3(\text{s}) \rightarrow \text{K}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$: $\Delta_{\text{LE}} H = -687 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H(\text{K}^+) = -322$, $\Delta_{\text{hyd}} H(\text{NO}_3^-) = -314$.

(a) Calculate $\Delta_{\text{sol}} H$ for KNO₃. [2]

(b) If $\Delta S_{\text{system}} = +115 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate ΔS_{total} at 298 K and explain why KNO₃ dissolves. [3]

Q37. 37. A* Challenge: Born-Haber Cycle for Silver Chloride AgCl**[5 Marks]**
WCH14/01/Hard/Q37

Data (kJ mol^{-1}): $\Delta_f H(\text{AgCl}) = -127$, $\Delta_{\text{at}} H(\text{Ag}) = +285$, $\text{IE}_1(\text{Ag}) = +731$, $\frac{1}{2} \Delta_{\text{at}} H(\text{Cl}_2) = +122$, $\text{EA}_1(\text{Cl}) = -349$.

(a) Calculate experimental Born-Haber $\Delta_{\text{LE}} H$ for AgCl. [3]

(b) The theoretical ionic lattice energy for AgCl is -770 kJ mol^{-1} . Account for the difference. [2]

Q38. 38. A* Challenge: Enthalpy of Atomisation of Metals vs Lattice Energy

[5 Marks]
WCH14/01/Hard/Q38

$\Delta_{\text{at}} H$ (kJ mol^{-1}): Na = +107, Mg = +148, Al = +326.

(a) Explain the trend in enthalpy of atomisation across Period 3 metals. [3]

(b) Relate metallic bond strength to charge density of metal cations and delocalised electron count. [2]

Q39. 39. A* Challenge: Born-Haber Cycle for Zinc Sulfide ZnS

[5 Marks]
WCH14/01/Hard/Q39

Data (kJ mol^{-1}): $\Delta_f H(\text{ZnS}) = -206$, $\Delta_{\text{at}} H(\text{Zn}) = +131$, $\text{IE}_1(\text{Zn}) + \text{IE}_2(\text{Zn}) = +2735$, $\Delta_{\text{at}} H(\text{S}) = +279$, $\text{EA}_1(\text{S}) + \text{EA}_2(\text{S}) = +332$.

(a) Calculate $\Delta_{\text{LE}} H$ for ZnS(s). [4]

(b) Explain why ZnS adopts a covalent zinc blende structure rather than a purely ionic NaCl lattice. [1]

Q40. 40. A* Challenge: Dissolution Energetics of Calcium Chloride CaCl₂

[5 Marks]
WCH14/01/Hard/Q40

$\text{CaCl}_2(\text{s}) \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$: $\Delta_{\text{LE}} H = -2258 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H(\text{Ca}^{2+}) = -1577$, $\Delta_{\text{hyd}} H(\text{Cl}^{-}) = -364$.

(a) Calculate $\Delta_{\text{sol}} H$ for CaCl₂. [3]

(b) Why is CaCl₂ used in self-heating food cans and de-icing roads? [2]

Q41. 41. A* Challenge: Polarization and Thermal Decomposition of Hydroxides

[5 Marks]
WCH14/01/Hard/Q41

$\text{Mg}(\text{OH})_2(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{H}_2\text{O}(\text{g})$ occurs at 350 °C, whereas Ba(OH)₂ melts at 780 °C without decomposing.

(a) Explain in terms of polarizing power of Mg²⁺ vs Ba²⁺ why Mg(OH)₂ decomposes easily. [5]

Q42. 42. A* Challenge: Enthalpy of Hydration of Gaseous Protons $\text{H}^+(\text{g})$ **[5 Marks]**
WCH14/01/Hard/Q42

The hydration enthalpy of $\text{H}^+(\text{g})$ is exceptionally exothermic: $\Delta_{\text{hyd}} \text{H}(\text{H}^+) = -1090 \text{ kJ mol}^{-1}$.

- (a) Explain why $\text{H}^+(\text{g})$ has a vastly more exothermic hydration enthalpy than $\text{Na}^+(\text{g})$ (-406 kJ mol^{-1}). [3]

- (b) Write the chemical equation for $\text{H}^+(\text{g})$ hydration forming $\text{H}_3\text{O}^+(\text{aq})$. [2]

Q43. 43. A* Challenge: Born-Haber Cycle for Sodium Fluoride NaF**[5 Marks]**
WCH14/01/Hard/Q43

Data (kJ mol^{-1}): $\Delta_{\text{f}} \text{H}(\text{NaF}) = -574$, $\Delta_{\text{at}} \text{H}(\text{Na}) = +107$, $\text{IEI}(\text{Na}) = +496$, $\frac{1}{2} \Delta_{\text{at}} \text{H}(\text{F}_2) = +79$, $\text{EA1}(\text{F}) = -328$.

- (a) Calculate $\Delta_{\text{LE}} \text{H}$ for $\text{NaF}(\text{s})$. [4]

- (b) Compare $\Delta_{\text{LE}} \text{H}(\text{NaF})$ with $\Delta_{\text{LE}} \text{H}(\text{NaCl})$ (-787 kJ mol^{-1}). [1]

Q44. 44. A* Challenge: Solvation Enthalpies in Non-Aqueous Solvents**[5 Marks]**
WCH14/01/Hard/Q44

Delta_{sol} H(LiCl) in liquid ammonia is -120 kJ mol⁻¹ compared to -37 kJ mol⁻¹ in water.

- (a) Explain why solvation enthalpy of Li⁺ is more exothermic in ammonia than in water. [3]

- (b) Relate nitrogen lone pair electron-donating ability to oxygen in water. [2]

Q45. 45. A* Challenge: Theoretical vs Experimental Lattice Energies of Transition Metal Halides**[5 Marks]**
WCH14/01/Hard/Q45

Delta_{LE} H (kJ mol⁻¹): MnCl₂ (exp -2520, theo -2480); FeCl₂ (exp -2530, theo -2490); CuCl₂ (exp -2750, theo -2580).

- (a) Explain why CuCl₂ shows a much larger discrepancy (170 kJ mol⁻¹) than MnCl₂ (40 kJ mol⁻¹). [3]

- (b) Relate this to Crystal Field Stabilization Energy (CFSE) and Jahn-Teller distortion. [2]

Q46. 46. A* Challenge: Born-Haber Cycle for Lead(II) Chloride PbCl₂**[5 Marks]**
WCH14/01/Hard/Q46

Data (kJ mol⁻¹): Delta_f H(PbCl₂) = -359, Delta_{at} H(Pb) = +195, IE₁(Pb) + IE₂(Pb) = +2190, Delta_{at} H(Cl₂) = +244, 2 x EA₁(Cl) = -698.

- (a) Calculate experimental Delta_{LE} H for PbCl₂(s). [4]

(b) Why is PbCl_2 insoluble in cold water? [1]

Q47. 47. A* Challenge: Calculating Enthalpy of Solution from Calorimetry Data

[5 Marks]
WCH14/01/Hard/Q47

5.00 g of anhydrous CaCl_2 ($M_r = 111.0$) was dissolved in 100 cm³ of water in a calorimeter. Temperature rose by 8.4 °C.

(Specific heat capacity $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

(a) Calculate $q = m c \Delta T$ in Joules. [2]

(b) Calculate moles of CaCl_2 and deduce $\Delta_{\text{sol}} H$ in kJ mol⁻¹. [3]

Q48. 48. A* Challenge: Born-Haber Cycle for Potassium Iodide KI

[5 Marks]
WCH14/01/Hard/Q48

Data (kJ mol⁻¹): $\Delta_f H(\text{KI}) = -328$, $\Delta_{\text{at}} H(\text{K}) = +89$, $IE_1(\text{K}) = +418$, $\frac{1}{2} \Delta_{\text{at}} H(\text{I}_2) = +107$, $EA_1(\text{I}) = -295$.

(a) Calculate $\Delta_{\text{LE}} H$ for $\text{KI}(\text{s})$. [4]

(b) Why does KI have a theoretical lattice energy almost identical to experimental? [1]

Q49. 49. A* Challenge: Hydration Enthalpy vs Ionic Size Grid Analysis

[5 Marks]
WCH14/01/Hard/Q49

Predict and rank $\Delta_{\text{hyd}} H^\circ$ for Na^+ , K^+ , Mg^{2+} , Ca^{2+} .

(a) Rank the four ions in order of increasingly exothermic hydration enthalpy with full justification. [5]

Q50. 50. A* Challenge: Complete Energetics Master Synthesis

[6 Marks]
WCH14/01/Hard/Q50

A student investigates anhydrous $\text{MgF}_2(\text{s})$.

Data (kJ mol^{-1}): $\Delta_f H(\text{MgF}_2) = -1124$, $\Delta_{\text{at}} H(\text{Mg}) = +148$, $\text{IE}_1(\text{Mg}) + \text{IE}_2(\text{Mg}) = +2189$, $\Delta_{\text{at}} H(\text{F}_2) = +158$, $\text{EA}_1(\text{F}) = -328$.

(a) Calculate $\Delta_{\text{LE}} H$ for $\text{MgF}_2(\text{s})$. [3]

(b) Given $\Delta_{\text{hyd}} H(\text{Mg}^{2+}) = -1920$ and $\Delta_{\text{hyd}} H(\text{F}^-) = -506 \text{ kJ mol}^{-1}$, calculate $\Delta_{\text{sol}} H$ for MgF_2 . [3]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 12B

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Lattice Energy Definition 1. (a) Enthalpy change when one mole of an ionic solid is formed from its gaseous ions under standard conditions (2). 1. (b) Bond formation between oppositely charged gaseous ions is an exothermic process (1).	[3]
Q2	Q2: 2. Born-Haber Cycle Calculation for NaCl 2. (a) $\Delta_f H = \Delta_{at} H(\text{Na}) + IE1(\text{Na}) + 1/2 \Delta_{at} H(\text{Cl}_2) + EA1(\text{Cl}) + \Delta_{LE} H$ (1). $-411 = +107 + 496 + 122 - 349 + \Delta_{LE} H = +376 + \Delta_{LE} H$ (2). $\Delta_{LE} H = -411 - 376 = -787 \text{ kJ mol}^{-1}$ (1). 2. (b) More exothermic lattice energy gives higher melting point (1).	[5]
Q3	Q3: 3. First Electron Affinity Definition & Sign 3. (a) Enthalpy change when one mole of electrons is added to one mole of gaseous atoms to form one mole of gaseous I^- ions (2). 3. (b) Attracting incoming electron to positively charged nucleus releases energy (1).	[3]
Q4	Q4: 4. Second Electron Affinity Endothermic Nature 4. (a) $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$ (1). 4. (b) An electron is being added to a negatively charged O^- ion; energy is required to overcome electrostatic repulsion (2).	[3]
Q5	Q5: 5. Lattice Energy Comparison: NaCl vs MgCl_2 5. (a) Mg^{2+} has a higher ionic charge (+2) than Na^+ (+1) (1). Mg^{2+} has a smaller ionic radius than Na^+ (1). Mg^{2+} has a much higher charge density (1). Greater electrostatic attraction between Mg^{2+} and Cl^- ions (1).	[4]
Q6	Q6: 6. Experimental vs Theoretical Lattice Energies: Polarization & Covalency 6. (a) Ag^+ has high charge density and polarizes the large, highly polarizable I^- electron cloud (2). This distorts the electron cloud, introducing significant covalent character (1). Covalent bonds provide additional bonding strength beyond ionic attraction (1). 6. (b) Ag^+ is polarizing cation; I^- is polarizable anion (1).	[5]
Q7	Q7: 7. Fajans' Rules for Ionic Polarization 7. (a) Cation: Small ionic radius (1) and high ionic charge (high charge density) (1). Anion: Large ionic radius (1) and high ionic charge (1).	[4]
Q8	Q8: 8. Enthalpy Change of Solution & Hydration Cycle 8. (a) Enthalpy cycle: $\text{MgSO}_4(\text{s}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{SO}_4^{2-}(\text{g}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$ (2). 8. (b) $\Delta_{\text{sol}} H = \text{Sum } \Delta_{\text{hyd}} H - \Delta_{LE} H = (-1920 - 1041) - (-2870) = -2961 - (-2870) = -91 \text{ kJ mol}^{-1}$ (3).	[5]
Q9	Q9: 9. Solubility Trend of Group 2 Sulfates 9. (a) Cation radius increases down Group 2 (Mg^{2+} to Ba^{2+}) (1). Both lattice energy and hydration enthalpy become less exothermic down group (1). Sulfate ion SO_4^{2-} is very large, so lattice energy decreases only slightly (1). Hydration enthalpy of cation decreases significantly due to larger radius/lower charge density (1). $\Delta_{\text{sol}} H$ becomes more endothermic (less negative), reducing solubility (1).	[5]
Q10	Q10: 10. Born-Haber Cycle Calculation for MgCl_2 10. (a) $\Delta_f H = \text{Sum atomisation}/IE/EA + \Delta_{LE} H$ (1). $-642 = +148 + 738 + 1451 + 244 - 698 + \Delta_{LE} H = +1883 + \Delta_{LE} H$ (2). $\Delta_{LE} H = -642 - 1883 = -2525 \text{ kJ mol}^{-1}$ (1). 10. (b) Mg^{2+} has higher charge (+2 vs +1) and smaller radius, leading to much stronger lattice (1).	[5]
Q11	Q11: 11. Solubility Trend of Group 2 Hydroxides 11. (a) OH^- is a small anion compared to SO_4^{2-} (1). Down Group 2, as cation radius increases, lattice energy decreases significantly (1). Hydration enthalpy of cation also decreases (1). Decrease in lattice energy dominates over decrease in hydration enthalpy (1). $\Delta_{\text{sol}} H$ becomes more exothermic / less endothermic down group, increasing solubility (1).	[5]

Q12	Q12: 12. Enthalpy of Hydration Definition & Trend 12. (a) Enthalpy change when one mole of gaseous ions is dissolved in water to form an infinitely dilute solution under standard conditions (2). 12. (b) Charge increases (+1 to +3) and ionic radius decreases (1), increasing charge density and ion-dipole attraction to water molecules (1).	[4]
Q13	Q13: 13. Enthalpy of Atomisation Definition 13. (a) Enthalpy change when one mole of gaseous atoms is formed from an element in its standard state under standard conditions (2). 13. (b) $\Delta_{\text{at}} \text{H}(\text{Cl}_2)$ is half the bond dissociation enthalpy of $\text{Cl}-\text{Cl}$ (1).	[3]
Q14	Q14: 14. Theoretical Lattice Energy Calculation via Born-Landé Equation 14. (a) 1. Ions are 100% spherical (1). 2. Bonding is 100% ionic with zero electron cloud distortion / zero covalent character (1). 14. (b) Na^+ is non-polarizing so NaCl is almost 100% ionic (1). Ag^+ is strongly polarizing, inducing significant covalent character in AgCl (1).	[4]
Q15	Q15: 15. Born-Haber Cycle for Calcium Fluoride CaF_2 15. (a) $-1220 = +178 + 590 + 1145 + 158 - 656 + \Delta_{\text{LE}} \text{H} = +1415 + \Delta_{\text{LE}} \text{H}$ (3). $\Delta_{\text{LE}} \text{H} = -1220 - 1415 = -2635 \text{ kJ mol}^{-1}$ (1). 15. (b) Ca^{2+} requires 2 F^- ions for charge neutrality (1).	[5]
Q16	Q16: 16. Enthalpy of Solution of Sodium Hydroxide 16. (a) $\Delta_{\text{sol}} \text{H} = (-406 - 539) - (-900) = -945 - (-900) = -45 \text{ kJ mol}^{-1}$ (3). 16. (b) Reaction is exothermic ($\Delta_{\text{sol}} \text{H} < 0$), releasing heat to the solution (1).	[4]
Q17	Q17: 17. Born-Haber Cycle for Potassium Oxide K_2O 17. (a) Sum atomisation/IE/EA = $+178 + 836 + 249 - 141 + 798 = +1920 \text{ kJ mol}^{-1}$ (3). $-363 = 1920 + \Delta_{\text{LE}} \text{H} \Rightarrow \Delta_{\text{LE}} \text{H} = -363 - 1920 = -2283 \text{ kJ mol}^{-1}$ (2).	[5]
Q18	Q18: 18. Polarizing Power Comparison: Li^+ vs K^+ 18. (a) Li^+ has a much smaller ionic radius than K^+ (1). Li^+ has higher charge density and higher polarizing power (1). Li^+ distorts the electron cloud of I^- significantly more, introducing covalent character (1).	[3]
Q19	Q19: 19. Enthalpy of Hydration of Halide Ions 19. (a) Halide ionic radius increases down group ($\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$) (1). Charge density decreases down group (1). Electrostatic attraction between halide ion and δ^+ hydrogen of water decreases (1). Hydration enthalpy becomes less exothermic (less negative) (1).	[4]
Q20	Q20: 20. Born-Haber Cycle for Aluminum Oxide Al_2O_3 20. (a) Al^{3+} has high charge (+3) and small radius (1). O^{2-} has high charge (-2) and small radius (1). Product of ionic charges is $3 \times 2 = 6$ for Al_2O_3 vs $1 \times 1 = 1$ for NaCl (1). Electrostatic attraction between Al^{3+} and O^{2-} is immensely stronger (1). High charge density leads to extremely compact, stable lattice (1).	[5]
Q21	Q21: 21. Thermal Decomposition of Group 2 Nitrates 21. (a) Mg^{2+} is smaller than Ba^{2+} with higher charge density (1). Mg^{2+} polarizes the nitrate $\text{N}-\text{O}$ bond more strongly (1). Weakens the $\text{N}-\text{O}$ bond in nitrate ion (1). Requires less thermal energy to break bond and decompose (1).	[4]
Q22	Q22: 22. Solubility of Calcium Hydroxide vs Calcium Sulfate 22. (a) SO_4^{2-} is much larger than OH^- (1). CaSO_4 has larger lattice energy relative to hydration enthalpy (1). $\Delta_{\text{sol}} \text{H}$ for CaSO_4 is more endothermic (1). $\text{Ca}(\text{OH})_2$ has less endothermic $\Delta_{\text{sol}} \text{H}$, allowing greater dissolution (1).	[4]
Q23	Q23: 23. Born-Haber Cycle for Lithium Iodide LiI 23. (a) $-270 = +159 + 520 + 107 - 295 + \Delta_{\text{LE}} \text{H} = +491 + \Delta_{\text{LE}} \text{H}$ (3). $\Delta_{\text{LE}} \text{H} = -270 - 491 = -761 \text{ kJ mol}^{-1}$ (1). 23. (b) Significant covalent character allows dissolution in non-polar/polar organic solvents (1).	[5]
Q24	Q24: 24. Enthalpy of Hydration vs Lattice Energy for BaCl_2 24. (a) $\Delta_{\text{sol}} \text{H} = (-1305 + 2(-364)) - (-2056) = -2033 - (-2056) = -23 \text{ kJ mol}^{-1}$ (3). 24. (b) Exothermic ($\Delta_{\text{sol}} \text{H} < 0$) (1).	[4]

Q25	Q25: 25. Born-Haber Cycle for Copper(II) Oxide CuO 25. (a) $\text{Sum} = +338 + 746 + 1958 + 249 + 657 = +3948 \text{ kJ mol}^{-1}$ (3). $-157 = 3948 + \Delta_{\text{LE H}} \Rightarrow \Delta_{\text{LE H}} = -157 - 3948 = -4105 \text{ kJ mol}^{-1}$ (2).	[5]
Q26	Q26: 26. A* Challenge: Born-Haber Cycle with Unknown Electron Affinity 26. (a) $-524 = 178 + 590 + 1145 + 112 + 2(\text{EA1}(\text{Br})) - 2150 = 2225 - 2150 + 2(\text{EA1}(\text{Br})) = +75 + 2(\text{EA1}(\text{Br}))$ (3). $2(\text{EA1}(\text{Br})) = -524 - 75 = -599 \Rightarrow \text{EA1}(\text{Br}) = -299.5 \text{ kJ mol}^{-1}$ (2).	[5]
Q27	Q27: 27. A* Challenge: Percent Covalent Character from Lattice Energy Comparison 27. (a) $\% \text{ diff} = [(-905 - (-758)) / -758] \times 100\% = (-147 / -758) \times 100\% = 19.4\%$ (2). 27. (b) Ag^+ has 4d10 outer shell with incomplete shielding, high polarizing power (1). Large polarizable Br^- electron cloud is distorted (1). Introduces substantial covalent orbital overlap, adding covalent bond energy to ionic attraction (1).	[5]
Q28	Q28: 28. A* Challenge: Enthalpy of Solution Temperature Dependence & Entropy 28. (a) $\Delta_{\text{sol H}} = (-322 - 293) - (-649) = -615 - (-649) = +34 \text{ kJ mol}^{-1}$ (2). 28. (b) Dissolution is endothermic ($\Delta_{\text{sol H}} > 0$). Increasing temperature shifts equilibrium $\text{KI(s)} \rightleftharpoons \text{K}^+(\text{aq}) + \text{I}^-(\text{aq})$ to the right (1). $\Delta S_{\text{surroundings}}$ becomes less negative, making ΔS_{total} more positive, increasing solubility (2).	[5]
Q29	Q29: 29. A* Challenge: Lattice Energy of Hypothetical NaCl2 29. (a) $\sim -2500 \text{ kJ mol}^{-1}$ (1). 29. (b) $\text{IE2}(\text{Na})$ removes an electron from an inner 2p shell (noble gas core) requiring massive energy ($+4562 \text{ kJ mol}^{-1}$) (2). This huge energy input outweighs the extra lattice energy gained by Na^{2+} (1). 29. (c) $\Delta_{\text{f H}}(\text{NaCl2})$ is highly endothermic ($+200$ to $+400 \text{ kJ mol}^{-1}$) whereas $\Delta_{\text{f H}}(\text{NaCl})$ is exothermic (-411 kJ mol^{-1}). NaCl is thermodynamically stable (2).	[6]
Q30	Q30: 30. A* Challenge: Hydration Enthalpy vs Lattice Energy Competition in Halides 30. (a) As halide radius increases ($\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$), both lattice energy and hydration enthalpy become less exothermic (2). Lattice energy decreases much faster because $r_{\text{Li}^+} + r_{\text{halide}}$ decreases proportionally more for lattice packing (2). Decreasing lattice energy dominates over decreasing hydration enthalpy, making $\Delta_{\text{sol H}}$ increasingly exothermic (1).	[5]
Q31	Q31: 31. A* Challenge: Born-Haber Cycle for Calcium Oxide CaO 31. (a) $\text{Sum atom/IE/EA} = +178 + 590 + 1145 + 249 - 141 + 798 = +2819 \text{ kJ mol}^{-1}$ (2). $-635 = 2819 + \Delta_{\text{LE H}} \Rightarrow \Delta_{\text{LE H}} = -635 - 2819 = -3454 \text{ kJ mol}^{-1}$ (2). 31. (b) $2+$ and $2-$ charges in CaO give 4x electrostatic attraction force compared to $1+$ and $1-$ in NaCl (1).	[5]
Q32	Q32: 32. A* Challenge: Polarization Effect on Thermal Stability of Carbonates 32. (a) Na^+ has $+1$ charge (lower charge density) so it does not polarize the carbonate ion significantly (2). Ca^{2+} and Mg^{2+} have $+2$ charges and polarize carbonate C-O bond, weakening it (1). 32. (b) Mg^{2+} is smaller than Ca^{2+} so it has higher charge density and stronger polarizing power, weakening C-O bond more (2).	[5]
Q33	Q33: 33. A* Challenge: Born-Haber Cycle for Barium Chloride BaCl2 33. (a) $\text{Sum} = 180 + 503 + 965 + 244 - 698 = +1194$ (2). $-858 = 1194 + \Delta_{\text{LE H}} \Rightarrow \Delta_{\text{LE H}} = -858 - 1194 = -2052 \text{ kJ mol}^{-1}$ (2). 33. (b) Ba^{2+} has a larger ionic radius than Mg^{2+} , leading to weaker electrostatic attraction and less exothermic lattice energy (1).	[5]
Q34	Q34: 34. A* Challenge: Hydration Enthalpy vs Ion Size in Transition Metal Ions 34. (a) Fe^{3+} has higher charge ($+3$ vs $+2$) and smaller ionic radius ($0.64 \text{ \AA}$ vs $0.78 \text{ \AA}$) (1). Fe^{3+} has much higher charge density, pulling water oxygen lone pairs much more strongly (2). 34. (b) Fe^{3+} polarizes O-H bonds of attached water ligand molecules more strongly, weakening O-H bonds and releasing H^+ ions readily ($\text{pH} \sim 2$ vs $\text{pH} \sim 5$) (2).	[5]
Q35	Q35: 35. A* Challenge: Enthalpy Cycle for Anhydrous vs Hydrated Copper Sulfate 35. (a) $\Delta_{\text{H hyd}} = \Delta_{\text{sol H}}(\text{CuSO4(s)}) - \Delta_{\text{sol H}}(\text{CuSO4.5H2O(s)})$ (2) = $-66.5 - (+11.7) = -78.2 \text{ kJ mol}^{-1}$ (2). 35. (b) Water dissolves the solid as it hydrates, making direct measurement of solid hydration impossible (1).	[5]

Q36	Q36: 36. A* Challenge: Enthalpy of Solution of Potassium Nitrate KNO ₃ 36. (a) $\Delta_{\text{sol}} H = (-322 - 314) - (-687) = -636 - (-687) = +51.0 \text{ kJ mol}^{-1}$ (2). 36. (b) $\Delta S_{\text{surr}} = -51000 / 298 = -171.1 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{total}} = +115 - 171.1 = -56.1 \text{ J K}^{-1} \text{ mol}^{-1}$ at 298 K. Dissolves at higher temperature as ΔS_{surr} becomes less negative (3).	[5]
Q37	Q37: 37. A* Challenge: Born-Haber Cycle for Silver Chloride AgCl 37. (a) $-127 = +285 + 731 + 122 - 349 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -127 - 789 = -916 \text{ kJ mol}^{-1}$ (3). 37. (b) Experimental value (-916) is 146 kJ mol ⁻¹ more exothermic than theoretical (-770) due to significant covalent character / polarization of Cl ⁻ by Ag ⁺ (2).	[5]
Q38	Q38: 38. A* Challenge: Enthalpy of Atomisation of Metals vs Lattice Energy 38. (a) Atomisation enthalpy increases from Na to Al (3). 38. (b) Cation charge increases (+1 to +3), ionic radius decreases, and number of delocalised electrons per atom increases (1 → 3), strengthening metallic bonding immensely (2).	[5]
Q39	Q39: 39. A* Challenge: Born-Haber Cycle for Zinc Sulfide ZnS 39. (a) $\text{Sum} = 131 + 2735 + 279 + 332 = +3477$ (2). $-206 = 3477 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -206 - 3477 = -3683 \text{ kJ mol}^{-1}$ (2). 39. (b) Zn ²⁺ strongly polarizes S ²⁻ (large polarizable anion), resulting in dominant covalent bonding character (1).	[5]
Q40	Q40: 40. A* Challenge: Dissolution Energetics of Calcium Chloride CaCl ₂ 40. (a) $\Delta_{\text{sol}} H = (-1577 + 2(-364)) - (-2258) = -2305 - (-2258) = -47 \text{ kJ mol}^{-1}$ (3). 40. (b) Dissolution is strongly exothermic ($\Delta_{\text{sol}} H = -47 \text{ kJ mol}^{-1}$), generating heat (1). Dissolved ions depress freezing point of water effectively (1).	[5]
Q41	Q41: 41. A* Challenge: Polarization and Thermal Decomposition of Hydroxides 41. (a) Mg ²⁺ has a small ionic radius and high charge density / high polarizing power (1). Mg ²⁺ attracts oxygen electron density of the OH ⁻ ion (1). Weakens the O-H bond and facilitates splitting off H ₂ O (1). Ba ²⁺ has a much larger radius and low charge density, so it cannot weaken O-H bond sufficiently to induce decomposition (2).	[5]
Q42	Q42: 42. A* Challenge: Enthalpy of Hydration of Gaseous Protons H ⁺ (g) 42. (a) H ⁺ is a bare proton with zero electron shell and extremely small radius (~10 ⁻¹⁵ m) (1). It has an infinitely high charge density (1), reacting covalently with water lone pair to form dative H ₃ O ⁺ (aq) (1). 42. (b) $\text{H}^+(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq})$ (2).	[5]
Q43	Q43: 43. A* Challenge: Born-Haber Cycle for Sodium Fluoride NaF 43. (a) $\text{Sum} = 107 + 496 + 79 - 328 = +354$ (2). $-574 = 354 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -574 - 354 = -928 \text{ kJ mol}^{-1}$ (2). 43. (b) F ⁻ is smaller than Cl ⁻ , allowing closer ionic approach and stronger electrostatic attraction (-928 vs -787) (1).	[5]
Q44	Q44: 44. A* Challenge: Solvation Enthalpies in Non-Aqueous Solvents 44. (a) NH ₃ has a less electronegative nitrogen atom which donates its lone pair more readily to Li ⁺ (1). Forms stronger dative covalent coordination bonds [Li(NH ₃) ₄] ⁺ (2). 44. (b) Nitrogen is less electronegative than oxygen, holding its lone pair less tightly and forming stronger dative bonds to cations (2).	[5]
Q45	Q45: 45. A* Challenge: Theoretical vs Experimental Lattice Energies of Transition Metal Halides 45. (a) Cu ²⁺ (d ⁹) experiences strong Jahn-Teller distortion and high polarizing power, introducing significant covalent character (3). 45. (b) CFSE provides additional thermodynamic stabilization beyond simple electrostatic ionic model for d ⁹ Cu ²⁺ compared to high-spin d ⁵ Mn ²⁺ (CFSE = 0) (2).	[5]
Q46	Q46: 46. A* Challenge: Born-Haber Cycle for Lead(II) Chloride PbCl ₂ 46. (a) $\text{Sum} = 195 + 2190 + 244 - 698 = +1931$ (2). $-359 = 1931 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -359 - 1931 = -2290 \text{ kJ mol}^{-1}$ (2). 46. (b) High covalent character and strong lattice energy make $\Delta_{\text{sol}} H$ endothermic (1).	[5]

Q47	Q47: 47. A* Challenge: Calculating Enthalpy of Solution from Calorimetry Data 47. (a) $q = 100 \times 4.18 \times 8.4 = 3511.2 \text{ J} = 3.511 \text{ kJ}$ (2). 47. (b) Moles = $5.00 / 111.0 = 0.04505 \text{ mol}$ (1). $\Delta_{\text{sol}} H = -3.511 / 0.04505 = -77.9 \text{ kJ mol}^{-1}$ (2).	[5]
Q48	Q48: 48. A* Challenge: Born-Haber Cycle for Potassium Iodide KI 48. (a) Sum = $89 + 418 + 107 - 295 = +319$ (2). $-328 = 319 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -328 - 319 = -647 \text{ kJ mol}^{-1}$ (2). 48. (b) K^+ has low charge density (+1, large radius) so it does not polarize I^- , resulting in almost 100% pure ionic bonding (1).	[5]
Q49	Q49: 49. A* Challenge: Hydration Enthalpy vs Ionic Size Grid Analysis 49. (a) $\text{K}^+ < \text{Na}^+ < \text{Ca}^{2+} < \text{Mg}^{2+}$ (1). K^+ is 1+ with largest radius (least exothermic) (1); Na^+ is 1+ with smaller radius (1); Ca^{2+} is 2+ with larger radius than Mg^{2+} (1); Mg^{2+} is 2+ with smallest radius (highest charge density, most exothermic $\Delta_{\text{hyd}} H = -1920 \text{ kJ mol}^{-1}$) (1).	[5]
Q50	Q50: 50. A* Challenge: Complete Energetics Master Synthesis 50. (a) Sum = $148 + 2189 + 158 - 2(328) = 2495 - 656 = +1839$ (1.5). $\Delta_{\text{LE}} H = -1124 - 1839 = -2963 \text{ kJ mol}^{-1}$ (1.5). 50. (b) $\Delta_{\text{sol}} H = (-1920 + 2(-506)) - (-2963) = (-1920 - 1012) + 2963 = -2932 + 2963 = +31.0 \text{ kJ mol}^{-1}$ (3).	[6]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 12B**FAQ #1: Definition of Lattice Energy Direction****Category: Definition Trap****COMMON EXAMINER TRAP / MISTAKE:**

Defining lattice energy as dissociation (endothermic) instead of formation (exothermic).

MODEL ANSWER / EXAMINER EXPECTATION:

Edexcel IAL defines lattice energy as the enthalpy change when ONE mole of an ionic crystal is FORMED from its gaseous ions (always EXOTHERMIC, negative sign).

FAQ #2: Multiplying Electron Affinity for Di-halides**Category: Stoichiometry Trap****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting to multiply EA1 by 2 for MX₂ salts like MgCl₂ or CaF₂.

MODEL ANSWER / EXAMINER EXPECTATION:

For MX₂ compounds, 2 moles of halide ions are formed from 2 moles of halogen atoms. You MUST multiply EA1 by 2 (e.g. $2 \times -349 = -698 \text{ kJ mol}^{-1}$).

FAQ #3: Second Ionisation Energy for Group 2 Metals**Category: Ionisation Trap****COMMON EXAMINER TRAP / MISTAKE:**

Including only IE1 for Group 2 metals in Born-Haber cycles.

MODEL ANSWER / EXAMINER EXPECTATION:

Group 2 metals form M²⁺ ions. The cycle MUST include both IE1 and IE2 (e.g. $\text{Mg} \rightarrow \text{Mg}^+ \rightarrow \text{Mg}^{2+}$).

FAQ #4: Sign of Second Electron Affinity**Category: EA2 Trap****COMMON EXAMINER TRAP / MISTAKE:**

Assuming EA2 is exothermic like EA1.

MODEL ANSWER / EXAMINER EXPECTATION:

EA1 is exothermic (attraction to neutral atom), but EA2 is ALWAYS ENDOTHERMIC (positive) due to electrostatic repulsion between incoming e⁻ and 1⁻ anion.

FAQ #5: Theoretical vs Experimental Discrepancy**Category: Covalency Trap****COMMON EXAMINER TRAP / MISTAKE:**

Attributing lattice energy discrepancies to experimental measurement error.

MODEL ANSWER / EXAMINER EXPECTATION:

Discrepancy occurs because theoretical model assumes 100% spherical ionic bonding. Experimental Born-Haber cycle includes COVALENT CHARACTER caused by cation polarization of anion.

FAQ #6: Fajans' Polarization Criteria**Category: Fajans Rules****COMMON EXAMINER TRAP / MISTAKE:**

Confusing which ion polarizes and which is polarized.

MODEL ANSWER / EXAMINER EXPECTATION:

The CATION polarizes (requires high charge, small radius). The ANION is polarized (requires high charge, large polarizable electron cloud).

FAQ #7: Hydration Enthalpy Sign in Solution Cycle**Category: Hess Cycle Trap****COMMON EXAMINER TRAP / MISTAKE:**

Subtracting hydration enthalpy instead of lattice energy in $\Delta_{\text{sol}} H$.

MODEL ANSWER / EXAMINER EXPECTATION:

$\Delta_{\text{sol}} H = \sum \Delta_{\text{hyd}} H(\text{ions}) - \Delta_{\text{LE}} H$. Hydration is exothermic (-), lattice formation is exothermic (-), so subtracting lattice formation subtracts a negative number.

FAQ #8: Group 2 Sulfate Solubility Trend**Category: Solubility Trend****COMMON EXAMINER TRAP / MISTAKE:**

Explaining sulfate solubility trend without emphasizing large sulfate ion size.

MODEL ANSWER / EXAMINER EXPECTATION:

SO_4^{2-} is very large, so lattice energy changes little down group. Decreasing hydration enthalpy of cation dominates, making $\Delta_{\text{sol}} H$ more endothermic down group (less soluble).

FAQ #9: Group 2 Hydroxide Solubility Trend**Category: Solubility Trend****COMMON EXAMINER TRAP / MISTAKE:**

Confusing hydroxide solubility trend with sulfate trend.

MODEL ANSWER / EXAMINER EXPECTATION:

OH^- is small, so lattice energy decreases sharply down group. Decreasing lattice energy dominates over decreasing hydration enthalpy, making $\Delta_{\text{sol}} H$ more exothermic down group (more soluble).

FAQ #10: Ionic Charge vs Radius Impact on Lattice Energy**Category: Lattice Comparison****COMMON EXAMINER TRAP / MISTAKE:**

Treating ionic radius as having equal weight to ionic charge.

MODEL ANSWER / EXAMINER EXPECTATION:

Ionic charge has a SQUARED effect on electrostatic attraction force ($q_1 \times q_2$), so doubling charge quadruples lattice energy, whereas changing radius has a smaller percentage impact.